The bacilli enters through URT and it remains confined to the site of entry where they multiply and produces diphtheria toxin

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The toxin causes local necrotic changes along with superficial inflammatory reaction. The necrosed epithelium together with fibrinous exudate, leucocytes, erthrocytes and bacteria consitute the greyish psedudomembrane, which is a characteristic feature of diphtheritic infection.

↓

The membrane extends from oropharynx to larynx and into trachae. The mechanical obstruction of the trachae by psedudomembrane can cause suffocation and death.

Flowchart 7.3: Localized effect of diphtheria toxin

(damage of multiple peripheral nerves), paralysis of palatine (the top part of the inside of the mouth) and ciliary muscles.

3. Degenerative changes in adrenal glands, kidney and liver may occur.

7.6.5 Laboratory Diagnosis

Specimen: Two swabs from the lesions are collected. One swab is used for smear preparationand other swab for inoculation on culturemedia.

Direct microscopy: Smears are stained with both Gram stain and Albert stain.

a. Gram Staining – Gram positive slender rods were observed.

b. Albert staining – Club shaped with metachromatic granules were observed.

Culture: The swabis inoculated on Loeffler's serum slope, after overnight incubation at 37°C, the plates were observed for characteristic colonies, which are identified by gram staining.

7.6.6 Prophylaxis

Diphtheria can be controlled by immunization. Three methods of immunization are available (Table 7.6).

7.6.7 Treatment

The specific treatment for diphtheria consists of administration of antitoxin with dose of 20,000–100, 000 units of ADS intramuscularly and antibiotic therapy using penicillin.

7.7 *Clostridium Tetani*

The genus *Clostridium* consists of anaerobic, spore forming Gram positive bacilli. The spores are wider than the bacterial bodies, giving the bacillus a swollen appearance resembling a spindle. The name *Clostridium* is derived from the word **'kluster' (a spindle**). Most species are saprophytes found in soil, water and decomposing plant and animal matter. Some of the pathogens are normal flora of intestinal tract of human and animals.

The genus *Clostridium* includes bacteria that causes 3 major diseases of human – **Tetanus, gas gangrene and food poisoning**. *Clostridium* pathogenicity is mainly due to production of a powerful exotoxin.

Clostridium of medical importance may be classified based on diseases they produce, which is given the Table (7.7).

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